

HONGXIANG ZHANG

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RESEARCH INTERESTS

My research focuses on reliable and efficient AI systems, with an emphasis on coding agents and agent memory. These agents often operate over large, evolving contexts that make it difficult to retain relevant knowledge and make reliable decisions. I improve how agents acquire and use information through memory construction, attention steering, and adaptive decoding. I apply these methods to code refinement, repository-level issue resolution, hallucination mitigation, and LLM-guided fuzzing.

EDUCATION

Purdue University Ph.D in Computer Science; Advised by Prof. Tianyi Zhang	August 2025 - Present
University of California, Davis M.Sc. in Computer Science	September 2022 - June 2025
Australian National University B.Eng. (Honours) in Computer Science	July 2020 - September 2022
Shandong University B.Eng. in Computer Science	September 2018 - June 2022

WORK EXPERIENCE

Research Assistant Purdue University, HCSS Lab West Lafayette, IN	August 2025 – Present
· Built an ontology-driven data extraction pipeline and knowledge graph for AI security as part of the NSF Proto-OKN project. · Responsible for developing the knowledge base and agent system for Eli Lilly’s Medicine Foundry, enabling AI agents to reason over complex manufacturing knowledge, identify potential issues, and provide traceable decision support.	

ACADEMIC EXPERIENCE

Enhancing Multi-Agent Communication through Attention Steering with Context Relevance	
<i>Hongxiang Zhang, Yuan Tian, Tianyi Zhang</i>	[arXiv]
[Under review] Multi-agent systems; Attention Alignment; Context Management	
· Proposed a training-free context management method that steers each agent’s attention toward relevant conversation history using temporal and spatial decay. · Improved multi-agent performance across five benchmarks by up to 7.64 absolute points while remaining robust as the number of agents and interaction rounds increases.	
FLARE: Fine-Grained Diagnostic Feedback for LLM Code Refinement	
<i>Yinsheng Yao, Hongxiang Zhang, Weixi Tong, Tianyi Zhang</i>	[arXiv]
[Under review] Code Refinement; Fault Localization; Test-time Scaling	
· Proposed an iterative code-refinement framework centered on a lightweight fault localizer that predicts line-level suspiciousness for targeted LLM refinement. · Improved LiveCodeBench and BigCodeBench performance across five LLMs, with up to 7.42% gain without candidate search and 8.50% average gain with top-k search.	

Attention-Aligned Reasoning for Large Language Models

Hongxiang Zhang, Yuan Tian, Tianyi Zhang

[arXiv]

[Under review] | LLM Reasoning; Attention Alignment; Prompt Engineering

- Proposed a reasoning-time alignment method that uses step-by-step attention alignment to keep LLMs focused on the prompt and intermediate goals.
- Outperformed existing prompting-based approaches and RL-trained SLMs.

Active Layer-Contrastive Decoding Reduces Hallucination in Large Language Model Generation

Hongxiang Zhang, Hao Chen, Muhao Chen, Tianyi Zhang

[arXiv]

[EMNLP 2025 Main] | Hallucination Mitigation; Decoding; Reinforcement Learning

- Proposed an RL-guided decoding strategy that actively contrasts model layers to suppress hallucinations.
- Achieved state-of-the-art hallucination reduction across five generation benchmarks.

LLAMAFUZZ: Large Language Model Enhanced Greybox Fuzzing

Hongxiang Zhang, Yuyang Rong, Yifeng He, Hao Chen

[arXiv]

[AST 2026] | LLM for Systems; Greybox Fuzzing; Software Testing

- Built LLAMAFUZZ, an LLM-enhanced greybox fuzzer that learns structured input formats and mutation strategies for binary and text targets.
- Improved real-world coverage and bug finding, identifying 47 unique bugs and increasing branch coverage by 27.19% over AFL++.